



Advanced Coding Challenge

Grid Volatility

Commercial Context

<https://github.com/cbfacademy/advanced-challenge-grid-volatility/blob/main/docs/CHALLENGE.md>

Technical Expectations

The core stack for this challenge is **Python and SQL**. We recommend using DuckDB or SQLite as the SQL engine.

Participants are encouraged to leverage modern AI tools (e.g., Cursor, GitHub Copilot, Claude) to accelerate data wrangling and model scaffolding. However, code must remain clean, maintainable, reproducible, and run seamlessly from a clean clone.

Participants will be given a pre-packaged 'Grid Volatility Dataset' to minimise setup time. The focus here will be on robust data validation, cleaning, and dynamic data synthesis. Participants will be asked about their data cleaning process, particularly regarding gaps, outliers, etc.

Note: A simple, clearly reasoned approach is preferred to a complex one that teams can't explain.

Required Deliverables

Submissions must directly respond to and resolve the challenge outlined. They must include:

1. `DECISION_MEMO.md`: a single-page executive brief detailing:
 - a. **Recommendation**: State the action or decision you would advise a trading desk to take based on your analysis. Explain why it is the preferred option and summarise the evidence supporting it.
 - b. **Trade-offs**: Describe the key compromises, risks, and limitations associated with your recommendation. Explain what benefits are gained, what is sacrificed, which assumptions underpin the decision, and any plausible alternatives that were rejected.
 - c. **Metrics and kill conditions**: how you'd measure success, and what would cause you to revisit the decision.

2. **GitHub Repo:** a reviewer should be able to run the program and reproduce your key outputs without manual steps. Please include a clear README.md file detailing how to run it and what to expect.

Template repo:

Access the pre-packaged template repo here:

<https://github.com/cbfacademy/advanced-challenge-grid-volatility>. The dataset is included as part of the codebase. The guidance is to **fork this repository** and continue development.

How to submit:

Raise a PR against

<https://github.com/cbfacademy/advanced-challenge-grid-volatility/tree/main> at **16:30**.

Please note that any commits made after this timestamp will not be considered and that late submissions may not be reviewed.

Build Day Timeline

Zoom link to join event:

<https://us02web.zoom.us/j/82071308479?pwd=tJlqnN7JqBpy4ebB23zbaEge6SPrFb.1>

Time	Activity	Comments
10:00	Attendees join session	
10:05	Welcome	Siobhan (CBF)
10:10	Opening remarks	Ethel (Shell)
10:20	Introduce challenge outline	<i>tbc</i> - Shell
10:30	Team formation and initial brainstorming	At this time, mentors will be supportive thinkers, helping to work through early thinking, query assumptions, and consider an approach.
11:00	Build	Mentors help to keep teams unstuck and work through blockers. Role is supportive and offers guidance rather than instruction.
13:30	<i>Comfort Break</i>	
13:55	Welcome back	Siobhan (CBF)
14:00	Build	Mentors help to keep teams unstuck and work through blockers. Role is supportive and offers guidance rather than instruction.
16:30	Feature freeze and final submission	Tanya (CBF) • Pitch Day reminders • Next steps • Thank yous